

NOTTINGHAM EAST MIDLANDS, United Kingdom

WMO: 034185

Lat: 52.831N

Lon: 1.328W

Elev: 93

StdP: 100.21

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-2.0	-5.9	2.3	-0.7	-4.8	2.6	-0.2	15.4	8.7	14.1	8.7	3.1	270	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.8	18.3	24.2	17.2	22.8	16.6	19.2	24.5	18.2	22.7	17.3	21.4	4.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.4	21.0	16.2	11.6	19.9	15.8	11.3	19.4	54.8	24.6	51.5	22.8	48.9	21.2	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	10.9	9.6	-5.8	30.1	1.8	2.1	-7.1	31.6	-8.2	32.8	-9.2	34.0	-10.5	35.5		
			-5.6	21.5	2.1	1.9	-7.1	22.9	-8.3	24.0	-9.5	25.0	-11.0	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	4.5	4.9	6.5	8.9	11.8	14.8	17.1	16.8	14.3	11.0	7.2	4.9
	DBStd	5.39	3.12	3.22	2.99	3.01	2.96	2.65	2.60	2.45	2.41	2.86	3.13	3.59
	HDD10.0	772	170	145	113	55	14	0	0	0	1	22	91	161
	HDD18.3	2985	428	377	366	283	202	110	57	61	124	228	333	417
	CDD10.0	868	1	2	5	22	71	145	219	212	129	53	8	2
	CDD18.3	41	0	0	0	0	1	5	18	14	2	1	0	0
	CDH23.3	298	0	0	0	1	7	48	140	89	13	2	0	0
	CDH26.7	53	0	0	0	0	0	7	27	18	1	0	0	0
Wind	WSAvg	4.7	5.5	5.4	5.2	4.6	4.5	4.2	4.1	4.2	4.3	4.8	4.9	5.2
Precipitation	PrecAvg	622	54	43	46	48	50	58	52	57	52	50	53	57
	PrecMax	1462	148	176	142	124	374	142	215	136	127	136	110	143
	PrecMin	446	13	6	7	7	11	8	4	6	7	8	24	10
	PrecStd	164	28	33	26	28	58	36	41	28	31	29	21	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.0	16.0	21.0	24.1	27.8	29.8	29.0	25.2	20.1	15.8	13.8
		MCWB	11.1	11.0	10.8	14.0	16.6	18.8	19.3	19.0	17.2	16.2	13.8	11.9
	2%	DB	11.2	12.0	14.0	17.9	21.8	24.2	26.9	26.0	22.0	17.8	14.1	12.8
		MCWB	9.5	9.7	10.3	12.2	15.3	16.7	18.2	18.4	16.5	14.8	12.4	11.2
	5%	DB	10.2	10.9	12.8	15.9	19.2	22.1	24.8	23.1	20.1	16.2	13.0	11.2
		MCWB	8.7	8.8	9.7	11.2	13.9	15.9	17.5	17.1	15.4	13.9	11.2	9.7
	10%	DB	9.1	9.8	11.1	14.0	17.1	20.1	22.7	21.8	18.8	15.1	11.9	10.1
		MCWB	8.0	8.1	8.5	10.2	12.8	15.1	16.7	16.5	14.6	13.0	10.3	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.7	12.2	14.7	17.5	19.8	20.8	21.3	18.7	16.6	14.0	12.4
		MCDB	12.8	13.4	14.7	19.5	22.1	26.0	26.2	26.7	22.7	19.2	15.4	13.6
	2%	WB	10.1	10.2	11.1	12.9	15.7	17.7	19.2	19.1	17.3	15.3	12.8	11.4
		MCDB	11.2	11.6	13.4	16.6	20.5	22.2	24.9	23.9	20.9	17.4	14.1	12.7
	5%	WB	9.1	9.1	10.1	11.6	14.6	16.6	18.2	18.0	16.1	14.4	11.6	10.3
		MCDB	10.1	10.6	12.0	15.2	18.6	20.8	23.0	22.0	19.0	16.2	12.9	11.3
	10%	WB	8.0	8.0	9.1	10.5	13.4	15.6	17.2	17.1	15.3	13.3	10.5	9.0
		MCDB	9.1	9.5	11.2	13.4	16.5	19.3	21.7	20.6	17.8	15.0	11.7	10.0
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.5	6.7	7.9	8.4	8.4	8.8	8.4	7.5	6.2	5.2	4.6
		MCDBR	5.6	6.2	8.3	10.3	11.6	12.0	12.4	10.9	9.7	6.8	5.4	5.3
		MCWBR	5.4	4.9	5.6	6.1	6.5	6.2	5.5	5.3	5.3	4.8	4.8	5.0
	5% WB	MCDBR	5.6	5.9	7.5	9.5	10.5	10.5	10.9	9.3	8.2	6.5	5.3	5.4
		MCWBR	5.5	5.1	5.6	6.1	6.3	6.3	5.9	5.2	5.4	5.1	5.0	5.3
Clear-Sky Solar Irradiance	taub	0.323	0.338	0.370	0.396	0.400	0.401	0.400	0.387	0.368	0.350	0.324	0.311	
	taud	2.348	2.383	2.290	2.227	2.256	2.280	2.286	2.348	2.402	2.432	2.403	2.328	
	Ebn at Noon	656	768	810	833	849	851	844	836	807	743	652	593	
	Edh at Noon	60	78	104	125	127	126	123	109	92	73	57	52	
All-Sky Solar Radiation	RadAvg	0.73	1.42	2.49	3.82	4.75	5.02	4.85	4.00	2.98	1.72	0.90	0.59	
	RadStd	0.07	0.17	0.30	0.43	0.31	0.49	0.52	0.28	0.27	0.15	0.10	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (88 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page